Maximum likelihood estimators (MLEs) with standard errors for (truncated) power laws

Yannik Ostermann

Abstract

This supplementary document has the aim to document and explain the theoretical background of the published code.

In the first section, it is explained what MLEs are, how to derive them, and how an error estimate can be made.
Further, two forms of truncated power laws are presented, one of them being a Schechter function, and it is discussed how a hypothesis test can be used to determine the best fit model.
In the second section, the MLEs are derived for the truncated power laws, as well as the functional form of their error estimates.

This work was initially aimed towards an astrophysical context, especially the IGIMF theory, but its results can be helpful to perform model fits on any power law or power law-like function with strong downturn towards the high-end.

1 Maximum likelihood estimators

The slope of power law-distributed data samples, where $f(x) \propto x^a$, is often determined by using histograms. This leads to the so-called binning bias.
This bias can be avoided by methods that don't rely on binning such as maximum likelihood estimators (MLEs).
MLEs have been shown to outperform linear or logarithmic binning in estimating power law exponents regarding variance and bias (White et al. 2008).
The following section serves as a short theoretical introduction for readers unfamiliar with maximum likelihood estimators and was written using the book of Lehmann & Casella (1998) and an MLE introduction by statistician Sherry Towers (to be found under this link), who used the books of Scott Long (1997) and King (1998).

A MLE works on the principle of the likelihood function.
Let $\mathcal{L}(\Theta|x_i) =: \mathcal{L}_i(\Theta)$ be the likelihood function for the discrete data point x_i and a model with m parameters $\Theta = \{\Theta_1, \Theta_2, ...\Theta_m\}$.

The likelihood for a specific model configuration $\hat{\Theta}$ at the point x_i is given by the probability density $p(x, \hat{\Theta}) = \frac{f(x,\hat{\Theta})}{\int_{x_{\min}}^{x_{\max}} f(x,\hat{\Theta})\,\mathrm{d}x}$ of the model function $f(x, \Theta)$ as

$$\mathcal{L}_i(\hat{\Theta}) = p(x_i, \hat{\Theta})\,.$$

For a sample of size n, the likelihood of the sample for the specific model configuration $\hat{\Theta}$ is given as the product of the individual likelihoods

$$\mathcal{L}(\hat{\Theta}) = \mathcal{L}_1(\hat{\Theta}) \times \mathcal{L}_2(\hat{\Theta}) ... \times \mathcal{L}_{n-1}(\hat{\Theta}) \times \mathcal{L}_n(\hat{\Theta}) = \prod_{i=1}^{n} \mathcal{L}_i(\hat{\Theta}) = \prod_{i=1}^{n} p(x_i, \hat{\Theta})\,.$$

The likelihood function can be simplified into the log-likelihood function by taking the natural logarithm on both sides:

$$\ln(\mathcal{L}(\hat{\Theta})) = \sum_{i=1}^{n} \ln(p(x_i, \hat{\Theta}))$$

The maximum of the likelihood function can be found by differentiating the (log-)likelihood function with respect to its model parameters Θ_i and setting the derivatives to zero:

$$\frac{\partial \ln(\mathcal{L}(\Theta))}{\partial \Theta_i} = 0$$

An error estimate can be made from the Fisher information I which can be approximated as the negative of the expectation value of the second derivative of the (log-)likelihood function evaluated at the ML estimate.

$$I(\Theta) = -\Big\langle \frac{\partial^2 \ln \mathcal{L}(\Theta)}{\partial \Theta^2} \Big\rangle \approx -\Big[\frac{\partial^2 \ln \mathcal{L}(\Theta)}{\partial \Theta^2}\Big]\Big|_{\Theta=\Theta_{\mathrm{Fit}}}\,,$$

with $\Theta_{\rm Fit}$ the set of best-fit parameters from the root finding. The last approximation is true for large n (Crawford et al. 1970). For single parameter models, the variance is given by the inverse of the Fisher information such that

$$\sigma^2(\Theta) = [I(\Theta)]^{-1} = \frac{1}{I(\Theta)}\,,$$

and the standard error of the ML estimate is simply given as

$$\sigma(\Theta) = \sqrt{\frac{1}{I(\Theta)}}\,.$$

For models with multiple parameters, the variance matrix is given by the inverse of the information matrix. In this case, the information matrix is given as the negative expectation value of the (log-)likelihood functions Hessian matrix. Then, the error estimates of the best-fit parameters are given by the square-root of the diagonal entries of the variance matrix

$$\mathrm{var}(\Theta) = [I(\Theta)]^{-1} = [-\langle H(\Theta)\rangle]^{-1} = \left[-\left\langle \frac{\partial^2 \ln \mathcal{L}(\Theta)}{\partial\Theta_i \partial\Theta_j}\right\rangle\right]^{-1} \approx \left[-\left(\frac{\partial^2 \ln \mathcal{L}(\Theta)}{\partial\Theta_i \partial\Theta_j}\right)\Big|_{\Theta=\Theta_{\rm Fit}}\right]^{-1}.$$

The last approximation is true for large data samples. To avoid confusion, note that here $I(\theta)$ refers to the Fisher information of the whole sample, opposed to the Fisher information of only a single data point, here $I_i(\theta)$.
The relationship between the quantities is: $I(\theta) = \sum_{i=1}^{n} I_i(\theta)$.

1.1 MLE for a pure power law

A ML estimator to determine the exponent of a power law was introduced first by Crawford et al. (1970). The following notation was published in Maschberger & Kroupa (2009). The slope of a power law distribution can be determined by a ML estimator by finding the root of

$$-\frac{n}{1-\hat{\alpha}_{\rm ML}} + n\frac{Z^{1-\hat{\alpha}_{\rm ML}}\ln(Z) - Y^{1-\hat{\alpha}_{\rm ML}}\ln(Y)}{Z^{1-\hat{\alpha}_{\rm ML}} - Y^{1-\hat{\alpha}_{\rm ML}}} - T = 0\,,$$

with n being the number of data points x_i, with $\hat{\alpha}_{\rm ML}$ the estimate on the slope, $Z = \max x_i$, $Y = \min x_i$ and $T = \sum_{i=1}^{n} \ln(x_i)$. The ML estimator $\hat{\alpha}_{\rm ML}$ can be found e.g. by using the `root_scalar` function implemented in Python.
In this work I use the modified ML estimator introduced by Maschberger & Kroupa (2009). It is similar to the ML estimator by Crawford, but the estimate on the slope $\hat{\alpha}$ is corrected for bias such that

$$\hat{\alpha}_{\rm MML} - 1 = \frac{n}{n-2}(\hat{\alpha}_{\rm ML} - 1)\,,$$

with $\hat{\alpha}_{\rm ML}$ being the slope determined with the ML estimator, $\hat{\alpha}_{\rm MML}$ the unbiased slope and n the number of data points.

1.2 Power laws with high-end downturns

Real life data samples may not follow a pure power law, but instead display a downturn at the high-end. Due to this, the data might not be well described by a power law fit function, as it may overestimate the steepness of the slope that is ought to be determined in the data analysis.
In order to account for the downturn, two additional fit models are used. The first is simply a Schechter function,

$$f(m) = km^{-\alpha} \mathrm{e}^{-\frac{m}{M_{\mathrm{turn}}}} \ .$$

The Schechter function is one of the simplest ways to model a power law downturn. It was shown e.g. in Lieberz & Kroupa (2017), that sums of power laws can lead to Schechter-like forms.
The second model is a modified version of the Schechter function that can account for a downturn that is even stronger than exponential. It will be called *modified* Schechter function(MS) from here on.

$$f(m) = km^{-\alpha} \mathrm{e}^{-(\frac{m}{M_{\mathrm{turn}}})^{\gamma}}$$

An example of a drawn distribution with the corresponding MLE fit functions is shown in Fig. 1. It displays a sample of summed power law distributions, using the IMF as the power law ($\xi(m) \propto m^{-\alpha}$), which is the number density distribution of stellar masses.

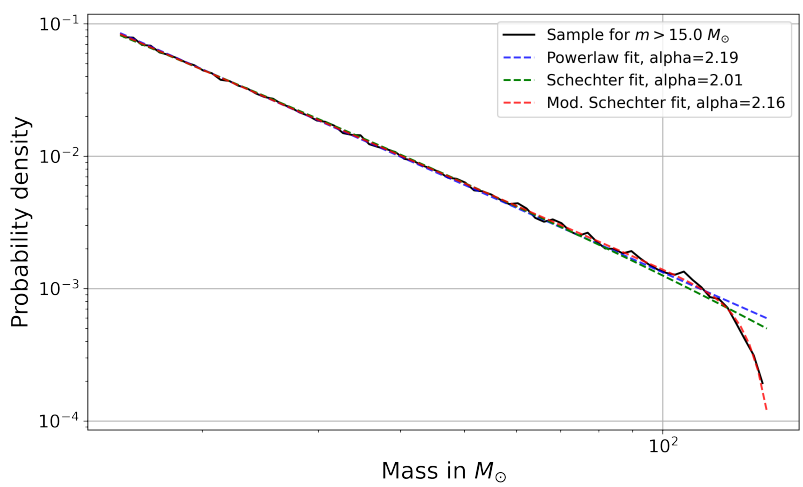

Figure 1: MLE fits to a generated sample of stellar masses. The sample represents a composite IMF, i.e. a sum of power law functions.

A rapid downturn of the sampled data at the high-mass end is evident and one can inspect by eye that neither a power law nor Schechter function provide an

adequate fit to the data. The modified Schechter function can be helpful in the context of the IGIMF theory to describe the downturn of composite IMFs and IGIMFs. To determine when the use of the more complicated models is justified over the simple models, a likelihood-ratio test is performed, which is explained in the next section.

1.3 Comparing model performance with the likelihood-ratio test

The likelihood-ratio test is a hypothesis test for nested models. Nested model means that the simple model can be found within the more complex model as a special case.
E.g. the simple model $f(x) = ax$ is nested within the more complex model $g(x) = ax^b$. For two models for which all parameters are fixed, so-called simple models, the likelihood-ratio test is the most powerful hypothesis test, as proven by the Neyman-Pearson lemma (Neyman & Pearson 1933). It will be used to determine which model provides the most accurate description without letting the fit get unnecessary complex.
A model with more parameters might yield a higher likelihood than a simpler model, but that does not necessarily make it a better model to the given data. The likelihood-ratio test serves to investigate whether the improvement in likelihood is significant enough to justify the use of the more complex model. To determine this significance one uses the likelihood-ratio statistic, which is defined as

$$\Lambda(x) = 2(\ln \mathcal{L}(x, \Theta) - \ln \mathcal{L}(x, \Theta_0)) \,,$$

with Θ_0 being part of the larger parameter space Θ and x the data.
Wilks' theorem (Wilks 1938) states that for a large number of data n the likelihood-ratio statistic converges to the χ^2_{d}-distribution ($\Lambda(x) \overset{n\to\infty}{\to} \chi^2_{\mathrm{d}}$) where d is the difference in the number of parameters or degrees of freedom, respectively ($\mathrm{d} = \dim(\Theta) - \dim(\Theta_0) > 0$). Therefore, one can use the χ^2_{d}-distribution in order to determine if the improvement in likelihood for an advanced model is significant. Since this is an approximation for large n, it should be considered whether the sample sizes used are large enough to justify this criterion.
For a specific value of the likelihood-ratio statistic $\hat{\Lambda}$ one can determine the statistical significance by determining the cumulative distribution function (CDF) of the χ^2_d-distribution in the interval $[0, \hat{\Lambda}]$. The probability that the improvement in likelihood, described by $\hat{\Lambda}$, is due to random chance is then given by

$$p = 1 - \mathrm{CDF}(\chi^2_{\mathrm{d}}(\hat{\Lambda})) \,.$$

Or in other words, the p-value will be the probability, that a random variable drawn from the χ^2_{d} distribution takes a value $\geq \hat{\Lambda}$ (see, for example Abdey 2013). If e.g. $p = 0.2$, this means that the probability that the more complex model fits better only due to random chance is 20%. As criterion to favor the results from the more advanced models it is often demanded that $p < 0.05$.

1.3.1 A short note on the validity of this method for optimally sampled data

It should be noted that there might be no appropriate criterion for a hypothesis test for optimally sampled data[1]. This is because optimally sampled data contain no statistical uncertainties by construction.
When combining various sets of optimally sampled data, as done for composite IMFs such as the IGIMF, something similar to random scatter may appear again as the form of the composite datasets is less confined compared to the form of singular optimally sampled datasets. This may imitate true random scatter.
How this affects the interpretation of the metrics of goodness-of-fit and hypothesis tests, which usually require randomness, is to my best knowledge not studied and more research regarding this is probably needed.
One can reason that the same p-value is probably a weaker criterion for composite optimally sampled datasets than for randomly sampled ones, since the decreased statistical noise leads to a very sharp form of the data which would generally increase the difference in likelihoods between simpler and more advanced models. That is, if there even appears some kind of randomness at all for composite optimally sampled datasets. If not, these tests cannot be applied to measure model performance.
For the moment, some way to make distinctions between models for composite sets of optimally sampled data is needed, and one may use the Likelihood-ratio test as a criterion to find the best functional description for a set of data, but one should keep in mind that using it gives no statistically correct results.

[1] Optimally sampled data are generated via the so-called *Optimal sampling* algorithm, which is again connected to the IGIMF theory.

2 Constructing MLE estimators and calculating error estimates

Maximum likelihood estimators outperform fitting methods using histogram bins by far in accuracy while being less biased.

Here, the mathematics are documented on how a Schechter and modified Schechter function ML function was constructed, as well as showing how the error estimates of those can be calculated, since the math is quite tedious. Note that some terms could be further simplified, e.g. by using the incomplete gamma function but for some programming languages, e.g. in `Python` this function is not well implemented for negative exponents in standard libraries or might have problems when root-finding algorithms are involved. Therefore, they are documented in this raw form.

2.1 Error estimate for the pure power law MLE

For a typical power law function

$$f(x) = kx^{-\alpha}\,,$$

the likelihood function for a dataset with size n is given by

$$\prod_{i=1}^{n} kx_i^{-\alpha} = k^n \prod_{i=1}^{n} x_i^{-\alpha} \quad \text{with} \quad k = \frac{1-\alpha}{x_{\max}^{1-\alpha} - x_{\min}^{1-\alpha}}\,,$$

where $x_{\min}$ and $x_{\max}$ are given by the minimum and maximum values of the dataset. The corresponding derivative of the log-likelihood function is

$$\frac{\partial \ln \mathcal{L}}{\partial \alpha} = -\frac{n}{1-\hat{\alpha}_{\mathrm{ML}}} + n\frac{Z^{1-\hat{\alpha}_{\mathrm{ML}}}\ln(Z) - Y^{1-\hat{\alpha}_{\mathrm{ML}}}\ln(Y)}{Z^{1-\hat{\alpha}_{\mathrm{ML}}} - Y^{1-\hat{\alpha}_{\mathrm{ML}}}} - T = 0\,,$$

with n the number of data points x_i, with $\hat{\alpha}_{\mathrm{ML}}$ the estimate on the slope, $Z = \max x_i$, $Y = \min x_i$ and $T = \sum_{i=1}^{n} \ln(x_i)$.
The error estimate of the slope can be determined by deriving the second derivative of the log-likelihood function with respect to α and plugging in the estimate value.

$$
\begin{aligned}
\frac{\partial^2 \ln \mathcal{L}}{\partial \alpha^2} &= n\frac{\partial}{\partial \alpha}\big(-\frac{1}{1-\alpha}\big)+n\frac{\partial}{\partial \alpha}\big(\frac{Z^{1-\alpha}\ln(Z)-Y^{1-\alpha}\ln(Y)}{Z^{1-\alpha}-Y^{1-\alpha}}\big) \\
&= -n\frac{1}{(1-\alpha)^2}+n\Big[\frac{\frac{\partial}{\partial\alpha}(Z^{1-\alpha}\ln(Z)-Y^{1-\alpha}\ln(Y))\cdot(Z^{1-\alpha}-Y^{1-\alpha})}{(Z^{1-\alpha}-Y^{1-\alpha})^2} \\
&- \frac{(Z^{1-\alpha}\ln(Z)-Y^{1-\alpha}\ln(Y))\cdot\frac{\partial}{\partial\alpha}(Z^{1-\alpha}-Y^{1-\alpha})}{(Z^{1-\alpha}-Y^{1-\alpha})^2}\Big] \\
&= -n\frac{1}{(1-\alpha)^2}+n\Big[\frac{((-\ln(Z)Z^{1-\alpha})\ln(Z)-(-\ln(Y)Y^{1-\alpha})\ln(Y))\cdot(Z^{1-\alpha}-Y^{1-\alpha})}{(Z^{1-\alpha}-Y^{1-\alpha})^2} \\
&- \frac{(Z^{1-\alpha}\ln(Z)-Y^{1-\alpha}\ln(Y))\cdot(-\ln(Z)Z^{1-\alpha}-(-\ln(Y))Y^{1-\alpha})}{(Z^{1-\alpha}-Y^{1-\alpha})^2}\Big] \\
&= -n\frac{1}{(1-\alpha)^2}+n\Big[\frac{(-\ln^2(Z)Z^{1-\alpha}+\ln^2(Y)Y^{1-\alpha})\cdot(Z^{1-\alpha}-Y^{1-\alpha})}{(Z^{1-\alpha}-Y^{1-\alpha})^2} \\
&- \frac{(Z^{1-\alpha}\ln(Z)-Y^{1-\alpha}\ln(Y))\cdot(-\ln(Z)Z^{1-\alpha}+\ln(Y)Y^{1-\alpha})}{(Z^{1-\alpha}-Y^{1-\alpha})^2}\Big] \\
&= -n\frac{1}{(1-\alpha)^2}+n\Big[\frac{-\ln^2(Z)(Z^{1-\alpha})^2+\ln^2(Z)Z^{1-\alpha}Y^{1-\alpha}+\ln^2(Y)Y^{1-\alpha}Z^{1-\alpha}-\ln^2(Y)(Y^{1-\alpha})^2}{(Z^{1-\alpha}-Y^{1-\alpha})^2} \\
&- \frac{-(Z^{1-\alpha})^2\ln^2(Z)+Z^{1-\alpha}\ln(Z)\ln(Y)Y^{1-\alpha}+Y^{1-\alpha}\ln(Y)\ln(Z)Z^{1-\alpha}-\ln^2(Y)(Y^{1-\alpha})^2}{(Z^{1-\alpha}-Y^{1-\alpha})^2}\Big] \\
&= -n\frac{1}{(1-\alpha)^2}+n\Big[\frac{\cancel{-\ln^2(Z)(Z^{1-\alpha})^2}+\ln^2(Z)Z^{1-\alpha}Y^{1-\alpha}+\ln^2(Y)Y^{1-\alpha}Z^{1-\alpha}\cancel{-\ln^2(Y)(Y^{1-\alpha})^2}}{(Z^{1-\alpha}-Y^{1-\alpha})^2} \\
&+ \frac{\cancel{(Z^{1-\alpha})^2\ln^2(Z)}-Z^{1-\alpha}\ln(Z)\ln(Y)Y^{1-\alpha}-Y^{1-\alpha}\ln(Y)\ln(Z)Z^{1-\alpha}\cancel{+\ln^2(Y)(Y^{1-\alpha})^2}}{(Z^{1-\alpha}-Y^{1-\alpha})^2}\Big] \\
&= -n\frac{1}{(1-\alpha)^2}+\frac{n}{(Z^{1-\alpha}-Y^{1-\alpha})^2}[\ln^2(Z)Z^{1-\alpha}Y^{1-\alpha}-2\ln(Y)\ln(Z)Y^{1-\alpha}Z^{1-\alpha}+\ln^2(Y)Y^{1-\alpha}Z^{1-\alpha}] \\
&= -n\frac{1}{(1-\alpha)^2}+\frac{n}{(Z^{1-\alpha}-Y^{1-\alpha})^2}Y^{1-\alpha}Z^{1-\alpha}[\ln^2(Z)-2\ln(Y)\ln(Z)+\ln^2(Y)] \\
&= -n\frac{1}{(1-\alpha)^2}+\frac{nY^{1-\alpha}Z^{1-\alpha}}{(Z^{1-\alpha}-Y^{1-\alpha})^2}\big(\ln(Z)-\ln(Y)\big)^2
\end{aligned}
$$

2.2 Creating a MLE for a Schechter function

The Schechter function is of the form

$$f(x) = kx^{-\alpha}\mathrm{e}^{-(\frac{x}{M_{\mathrm{turn}}})}\,,$$

The normalization constant k has to be chosen such that the likelihood function is normalized to 1.
The integral over the whole mass range is given as

$$\int_{m_{\mathrm{min}}}^{m_{\mathrm{max}}} f(x')\,\mathrm{d}x' \overset{!}{=} 1 \quad \Rightarrow k = \frac{1}{\int_{m_{\mathrm{min}}}^{m_{\mathrm{max}}} (x')^{-\alpha} \cdot \mathrm{e}^{-(\frac{x'}{M_{\mathrm{turn}}})}\,\mathrm{d}x'}\,.$$

The likelihood function for a specific data point x_i is dependent on the set of parameters

$$\Theta = \{\Theta_1, \Theta_2, ...\Theta_m\} \overset{m=2}{=} \{\alpha, M_{\mathrm{turn}}\}\,,$$

such that

$$\mathcal{L}(\alpha, M_{\mathrm{turn}}|x_i) = k(x_i)^{-\alpha} \cdot e^{-(\frac{x_i}{M_{\mathrm{turn}}})}\,.$$

And the corresponding log-likelihood is therefore:

$$\ln \mathcal{L}(\alpha, M_{\mathrm{turn}}|x_i) = \ln(k) - \alpha \ln(x_i) - (\frac{x_i}{M_{\mathrm{turn}}})\,.$$

For n datapoints the total likelihood function is the product of the likelihoods for each datapoint x_i

$$\mathcal{L} = k^n \Pi_{i=1}^n (x_i)^{-\alpha} \cdot \mathrm{e}^{-(\frac{x_i}{M_{\mathrm{turn}}})}\,.$$

The log-likelihood function for the whole dataset is then given as

$$\ln \mathcal{L} = n\ln(k) - \alpha \sum_{i=1}^{n} (\ln(x_i)) - \sum_{i=1}^{n} (\frac{x_i}{M_{\mathrm{turn}}})\,,$$
$$\Leftrightarrow \ln \mathcal{L} = n\ln(k) - \alpha \sum_{i=1}^{n} (\ln(x_i)) - M_{\mathrm{turn}}^{-1} \sum_{i=1}^{n} (x_i)\,.$$

In order to find the maximum likelihood, the derivatives of the log-likelihood function with respect to α and M_{turn} have to be found.
The derivative of the log-likelihood function with respect to α yields

$$\frac{\partial \ln \mathcal{L}}{\partial \alpha} = n\frac{\frac{\partial k}{\partial \alpha}}{k} - \sum_{i=1}^{n} (\ln(x_i))\,,$$

and the derivative with respect to M_{turn} is

$$\frac{\partial \ln \mathcal{L}}{\partial M_{\mathrm{turn}}} = n\frac{\frac{\partial k}{\partial M_{\mathrm{turn}}}}{k} + \frac{1}{M_{\mathrm{turn}}^2} \sum_{i=1}^{n} (x_i)\,.$$

The derivatives with respect to k are:

$$\begin{aligned}\frac{\partial k(\alpha)}{\partial \alpha} &= \frac{\partial}{\partial \alpha}\Big(\int_{m_{\min}}^{m_{\max}} (x')^{-\alpha} \cdot \mathrm{e}^{-(\frac{x}{M_{\mathrm{turn}}})}\,\mathrm{d}x'\Big)^{-1} \\ &= -\frac{1}{(\int_{m_{\min}}^{m_{\max}} (x')^{-\alpha} \cdot \mathrm{e}^{-(\frac{x'}{M_{\mathrm{turn}}})}\,\mathrm{d}x')^2}\frac{\partial}{\partial \alpha}\Big(\int_{m_{\min}}^{m_{\max}} (x')^{-\alpha} \cdot \mathrm{e}^{-(\frac{x'}{M_{\mathrm{turn}}})}\,\mathrm{d}x'\Big) \\ &= -k^2\frac{\partial}{\partial \alpha}\Big(\int_{m_{\min}}^{m_{\max}} (x')^{-\alpha} \cdot \mathrm{e}^{-(\frac{x'}{M_{\mathrm{turn}}})}\,\mathrm{d}x'\Big)\,.\end{aligned}$$

Since the function integrated over is continuously differentiable on the needed x interval one can interchange the derivative and integral operator.

$$\frac{\partial k(\alpha)}{\partial \alpha} = -k^2\int_{m_{\min}}^{m_{\max}} \frac{\partial}{\partial \alpha}(x')^{-\alpha} \cdot \mathrm{e}^{-(\frac{x'}{M_{\mathrm{turn}}})}\,\mathrm{d}x'\,.$$

From $(x')^{-\alpha} = \mathrm{e}^{-\alpha\ln(x')}$, follows that

$$\frac{\partial}{\partial \alpha}(x')^{-\alpha} = -\ln(x')\mathrm{e}^{-\alpha\ln(x')} = -\ln(x')(x')^{-\alpha}\,.$$

And from this follows that

$$\frac{\partial k(\alpha)}{\partial \alpha} = -k^2\int_{m_{\min}}^{m_{\max}} \Big(-\ln(x')(x')^{-\alpha}\Big)\cdot\mathrm{e}^{-(\frac{x'}{M_{\mathrm{turn}}})}\,\mathrm{d}x' = k^2\int_{m_{\min}}^{m_{\max}} \ln(x')(x')^{-\alpha} \cdot \mathrm{e}^{-(\frac{x'}{M_{\mathrm{turn}}})}\,\mathrm{d}x'\,.$$

The derivative of k by M_{turn} follows a similar manner. The derivative under the integral is determined via

$$\frac{\partial}{\partial M_{\mathrm{turn}}}(x')^{-\alpha} \cdot \mathrm{e}^{-(\frac{x}{M_{\mathrm{turn}}})} = (x')^{-\alpha}\frac{\partial}{\partial M_{\mathrm{turn}}}\mathrm{e}^{-(\frac{x}{M_{\mathrm{turn}}})}\,,$$

with $e^{-(\frac{x}{M_{\mathrm{turn}}})} = (\mathrm{e}^{(\frac{x}{M_{\mathrm{turn}}})})^{-1}$ from which follows

$$\frac{\partial}{\partial M_{\mathrm{turn}}}\mathrm{e}^{-(\frac{x}{M_{\mathrm{turn}}})} = \frac{\partial}{\partial M_{\mathrm{turn}}}(-\frac{x}{M_{\mathrm{turn}}})\mathrm{e}^{-(\frac{x}{M_{\mathrm{turn}}})} = \frac{x}{M_{\mathrm{turn}}^2}\mathrm{e}^{-(\frac{x}{M_{\mathrm{turn}}})}\,.$$

Therefore, the total term is

$$\frac{\partial k(M_{\mathrm{turn}})}{\partial M_{\mathrm{turn}}} = -k^2\int_{m_{\min}}^{m_{\max}} \frac{\partial}{\partial M_{\mathrm{turn}}}(x')^{-\alpha}\cdot\mathrm{e}^{-((\frac{x'}{M_{\mathrm{turn}}})^{\gamma})}\,\mathrm{d}x' = -k^2\int_{m_{\min}}^{m_{\max}} (x')^{-\alpha}\frac{x'}{M_{\mathrm{turn}}^2}\mathrm{e}^{-(\frac{x'}{M_{\mathrm{turn}}})}\,\mathrm{d}x'\,.$$

2.2.1 Error calculation for the Schechter function MLE

The second derivatives of the log-likelihood function needed to calculate an error estimate are:

$$\frac{\partial^2 \mathcal{L}}{\partial \alpha^2} = \frac{\partial}{\partial \alpha}\frac{\partial \ln \mathcal{L}}{\partial \alpha} = \frac{\partial}{\partial \alpha}(n\frac{\frac{\partial k}{\partial \alpha}}{k} - \sum_{i=1}^{n}(\ln(x_i))) = n\frac{\frac{\partial^2 k}{\partial \alpha^2}k - (\frac{\partial k}{\partial \alpha})^2}{k^2} - 0 = n\frac{\frac{\partial^2 k}{\partial \alpha^2}k - (\frac{\partial k}{\partial \alpha})^2}{k^2}\,,$$

$$\frac{\partial^2 \mathcal{L}}{\partial \alpha \partial M_{\text{turn}}} = \frac{\partial}{\partial M_{\text{turn}}}\frac{\partial \ln \mathcal{L}}{\partial \alpha} = \frac{\partial}{\partial M_{\text{turn}}}(n\frac{\frac{\partial k}{\partial \alpha}}{k} - \sum_{i=1}^{n}\ln(x_i)) = n\frac{\frac{\partial^2 k}{\partial \alpha \partial M_{\text{turn}}}k - \frac{\partial k}{\partial \alpha}\frac{\partial k}{\partial M_{\text{turn}}}}{k^2}\,,$$

with $\frac{\partial^2 \mathcal{L}}{\partial \alpha \partial M_{\text{turn}}} = \frac{\partial^2 \mathcal{L}}{\partial M_{\text{turn}} \partial \alpha}$ since the derivatives acted on k are interchangeable.
And lastly,

$$\frac{\partial^2 \mathcal{L}}{\partial M_{\text{turn}}^2} = \frac{\partial}{\partial M_{\text{turn}}}\frac{\partial \ln \mathcal{L}}{\partial M_{\text{turn}}} = \frac{\partial}{\partial M_{\text{turn}}}(n\frac{\frac{\partial k}{\partial M_{\text{turn}}}}{k} + \frac{1}{M_{\text{turn}}^2}\sum_{i=1}^{n}(x_i)) = n\frac{\frac{\partial^2 k}{\partial M_{\text{turn}}^2}k - (\frac{\partial k}{\partial M_{\text{turn}}})^2}{k^2} - \frac{2}{M_{\text{turn}}^3}\sum_{i=1}^{n}(x_i)\,.$$

The second derivatives of k, are

$$\frac{\partial^2 k}{\partial \alpha^2} = -k^2 \int_{m_{\min}}^{m_{\max}} (\ln(x'))^2 (x')^{-\alpha} \cdot \mathrm{e}^{-(\frac{x'}{M_{\text{turn}}})}\,\mathrm{d}x'$$

$$\begin{aligned}\frac{\partial^2 k}{\partial M_{\text{turn}}^2} &= k^2 \int_{m_{\min}}^{m_{\max}} (x')^{-\alpha} \cdot \mathrm{e}^{-(\frac{x'}{M_{\text{turn}}})}\Big(-\frac{x'(2M_{\text{turn}} - x')}{M_{\text{turn}}^4}\Big)\,\mathrm{d}x' \\ &= -k^2 \int_{m_{\min}}^{m_{\max}} (x')^{-\alpha} \cdot \mathrm{e}^{-(\frac{x'}{M_{\text{turn}}})}\frac{x'(2M_{\text{turn}} - x')}{M_{\text{turn}}^4}\,\mathrm{d}x'\end{aligned}$$

$$\begin{aligned}\frac{\partial^2 k}{\partial \alpha \partial M_{\text{turn}}} &= -k^2 \int_{m_{\min}}^{m_{\max}} \Big(-\ln(x')(x')^{-\alpha} \cdot \mathrm{e}^{-(\frac{x'}{M_{\text{turn}}})}\frac{x'}{M_{\text{turn}}^2}\Big)\,\mathrm{d}x' \\ &= k^2 \int_{m_{\min}}^{m_{\max}} \ln(x')(x')^{-\alpha} \cdot \mathrm{e}^{-(\frac{x'}{M_{\text{turn}}})}\frac{x'}{M_{\text{turn}}^2}\,\mathrm{d}x'\end{aligned}$$

2.3 Creating a MLE for a modified, Schechter-like function

For some samples, the rapid downturn in number cannot be described by a simple exponential downturn. Therefore, another function with a stronger than exponential decrease is constructed, that will in the following be called modified Schechter (MS) function.
The newly constructed function is of the form

$$f(x) = k(x)^{-\alpha} \frac{1}{\mathrm{e}^{(\frac{x}{M_{\mathrm{turn}}})^{\gamma}}} \,.$$

Again, the normalization constant k has to be chosen such that the likelihood is normalized to 1.

$$\int_{m_{\mathrm{min}}}^{m_{\mathrm{max}}} f(x')\,\mathrm{d}x' \stackrel{!}{=} 1 \Rightarrow \quad k = \frac{1}{\int_{m_{\mathrm{min}}}^{m_{\mathrm{max}}} (x')^{-\alpha} \cdot \mathrm{e}^{-((\frac{x}{M_{\mathrm{turn}}})^{\gamma})}\,\mathrm{d}x'} \,.$$

The Likelihood function for a specific data point x_i is dependent on the set of parameters

$$\Theta = \{\Theta_1, \Theta_2, ...\Theta_m\} \stackrel{m=3}{=} \{\alpha, M_{\mathrm{turn}}, \gamma\} \,,$$

such that

$$\mathcal{L}(\alpha, M_{\mathrm{turn}}, \gamma | x) = k(x)^{-\alpha} \cdot \mathrm{e}^{-((\frac{x}{M_{\mathrm{turn}}})^{\gamma})} \,.$$

The log-likelihood is therefore:

$$\ln \mathcal{L}(\alpha, M_{\mathrm{turn}}, \gamma | x) = \ln(k) - \alpha \ln(x) - (\frac{x}{M_{\mathrm{turn}}})^{\gamma} \,.$$

For n datapoints the total likelihood function is the sum of the log-likelihoods for each datapoint x_i

$$\mathcal{L} = k^n \Pi_{i=1}^{n} (x_i)^{-\alpha} \cdot \mathrm{e}^{-(\frac{x_i}{M_{\mathrm{turn}}})^{\gamma}} \,.$$

The log-likelihood function for the whole dataset is then

$$\ln \mathcal{L} = n \ln(k) - \alpha \sum_{i=1}^{n} (\ln(x_i)) - \sum_{i=1}^{n} (\frac{x_i}{M_{\mathrm{turn}}})^{\gamma}$$
$$\Leftrightarrow \ln \mathcal{L} = n \ln(k) - \alpha \sum_{i=1}^{n} (\ln(x_i)) - M_{\mathrm{turn}}^{-\gamma} \sum_{i=1}^{n} (x_i)^{\gamma}$$

If one differentiates the likelihood function by its parameters, this yields three equations that are set to zero in order to maximize the function.

$$\frac{\partial \ln \mathcal{L}}{\partial \alpha} = n\frac{\frac{\partial k(\alpha)}{\partial \alpha}}{k} - \sum_{i=1}^{n}(\ln(x_i))$$

$$\frac{\partial \ln \mathcal{L}}{\partial M_{\text{turn}}} = n\frac{\frac{\partial k(M_{\text{turn}})}{\partial M_{\text{turn}}}}{k} + \gamma M_{\text{turn}}^{-(\gamma+1)}\sum_{i=1}^{n}(x_i)^{\gamma}$$

$$\begin{aligned}\frac{\partial \ln \mathcal{L}}{\partial \gamma} &= n\frac{\frac{\partial k(\gamma)}{\partial \gamma}}{k} - \frac{\partial}{\partial \gamma}(\sum_{i=1}^{n}(\frac{x_i}{M_{\text{turn}}})^{\gamma}) = n\frac{\frac{\partial k(\gamma)}{\partial \gamma}}{k} - \frac{\partial}{\partial \gamma}(\sum_{i=1}^{n}\mathrm{e}^{\ln(\frac{x_i}{M_{\text{turn}}})\cdot\gamma}) \\ &= n\frac{\frac{\partial k(\gamma)}{\partial \gamma}}{k} - \sum_{i=1}^{n}(\ln(\frac{x_i}{M_{\text{turn}}}))\mathrm{e}^{\ln(\frac{x_i}{M_{\text{turn}}})\cdot\gamma} = n\frac{\frac{\partial k(\gamma)}{\partial \gamma}}{k} - \sum_{i=1}^{n}\ln(\frac{x_i}{M_{\text{turn}}})(\frac{x_i}{M_{\text{turn}}})^{\gamma}\end{aligned}$$

The derivatives with respect to k are determined as follows:

$$\begin{aligned}\frac{\partial k(\alpha)}{\partial \alpha} &= \frac{\partial}{\partial \alpha}(\int_{m_{\min}}^{m_{\max}}(x')^{-\alpha}\cdot \mathrm{e}^{-((\frac{x}{M_{\text{turn}}})^{\gamma})}\,\mathrm{d}x')^{-1} \\ &= \frac{-1}{(\int_{m_{\min}}^{m_{\max}}(x')^{-\alpha}\cdot \mathrm{e}^{-((\frac{x}{M_{\text{turn}}})^{\gamma})}\,\mathrm{d}x')^{2}}\frac{\partial}{\partial \alpha}(\int_{m_{\min}}^{m_{\max}}(x')^{-\alpha}\cdot \mathrm{e}^{-((\frac{x}{M_{\text{turn}}})^{\gamma})}\,\mathrm{d}x') \\ &= -k^2\frac{\partial}{\partial \alpha}(\int_{m_{\min}}^{m_{\max}}(x')^{-\alpha}\cdot \mathrm{e}^{-((\frac{x}{M_{\text{turn}}})^{\gamma})}\,\mathrm{d}x')\end{aligned}$$

Since the function under the integral is continuously differentiable on the needed x interval one can interchange the derivative and the integral operator.

$$\frac{\partial k(\alpha)}{\partial \alpha} = -k^2\int_{m_{\min}}^{m_{\max}}\frac{\partial}{\partial \alpha}(x')^{-\alpha}\cdot \mathrm{e}^{-((\frac{x}{M_{\text{turn}}})^{\gamma})}\,\mathrm{d}x'\,,$$

with $(x')^{-\alpha} = \mathrm{e}^{-\alpha\ln(x')}$, follows that

$$\frac{\partial}{\partial \alpha}(x')^{-\alpha} = -\ln(x')\mathrm{e}^{-\alpha\ln(x')} = -\ln(x')(x')^{-\alpha}\,.$$

And from this follows that

$$\frac{\partial k(\alpha)}{\partial \alpha} = -k^2\int_{m_{\min}}^{m_{\max}}(-\ln(x')(x')^{-\alpha})\cdot \mathrm{e}^{-((\frac{x}{M_{\text{turn}}})^{\gamma})}\,\mathrm{d}x' = k^2\int_{m_{\min}}^{m_{\max}}\ln(x')(x')^{-\alpha}\cdot \mathrm{e}^{-((\frac{x}{M_{\text{turn}}})^{\gamma})}\,\mathrm{d}x'\,.$$

The derivative of k with respect to M_{turn} follows in a similar manner such that

$$\frac{\partial}{\partial M_{\text{turn}}}(x')^{-\alpha}\cdot \mathrm{e}^{-((\frac{x}{M_{\text{turn}}})^{\gamma})} = (x')^{-\alpha}\frac{\partial}{\partial M_{\text{turn}}}\mathrm{e}^{-((\frac{x}{M_{\text{turn}}})^{\gamma})}\,,$$

with $\mathrm{e}^{-((\frac{x}{M_{\text{turn}}})^{\gamma})} = (\mathrm{e}^{(\frac{x}{M_{\text{turn}}})^{\gamma}})^{-1} = (\mathrm{e}^{\mathrm{e}^{\ln(\frac{x}{M_{\text{turn}}})^{\gamma}}})^{-1}$ from this follows

$$\begin{aligned}
\frac{\partial}{\partial M_{\text{turn}}}\mathrm{e}^{-((\frac{x}{M_{\text{turn}}})^{\gamma})} &= \frac{\partial}{\partial M_{\text{turn}}}(\mathrm{e}^{\mathrm{e}^{\ln(\frac{x}{M_{\text{turn}}})^{\gamma}}})^{-1}\\
&= -1(\mathrm{e}^{\mathrm{e}^{\ln(\frac{x}{M_{\text{turn}}})^{\gamma}}})^{-2}\frac{\partial}{\partial M_{\text{turn}}}\mathrm{e}^{\mathrm{e}^{\ln(\frac{x}{M_{\text{turn}}})^{\gamma}}}\\
&= \frac{-1}{(\mathrm{e}^{(\frac{x}{M_{\text{turn}}})^{\gamma}})^{\cancel{2}}}\cancel{\mathrm{e}^{(\frac{x}{M_{\text{turn}}})^{\gamma}}}\frac{\partial}{\partial M_{\text{turn}}}\mathrm{e}^{\ln(\frac{x}{M_{\text{turn}}})\gamma}\\
&= \frac{-1}{(\mathrm{e}^{(\frac{x}{M_{\text{turn}}})^{\gamma}})}\mathrm{e}^{\ln(\frac{x}{M_{\text{turn}}})\gamma}\gamma\frac{\partial}{\partial M_{\text{turn}}}(\ln(\frac{x}{M_{\text{turn}}}))\\
&= \frac{\cancel{-1}}{(\mathrm{e}^{(\frac{x}{M_{\text{turn}}})^{\gamma}})}(\frac{x}{M_{\text{turn}}})^{\gamma}\gamma\frac{\cancel{x}}{\cancel{M_{\text{turn}}}}\frac{\cancel{-1}}{M_{\text{turn}}^{\cancel{2}}}\\
&= \frac{\gamma(\frac{x'}{M_{\text{turn}}})^{\gamma}\mathrm{e}^{-(\frac{x'}{M_{\text{turn}}})^{\gamma}}}{M_{\text{turn}}}
\end{aligned}$$

Therefore the final term is

$$\frac{\partial k(M_{\text{turn}})}{\partial M_{\text{turn}}} = -k^2\int_{m_{\min}}^{m_{\max}}\frac{\partial}{\partial M_{\text{turn}}}(x')^{-\alpha}\cdot \mathrm{e}^{-((\frac{x}{M_{\text{turn}}})^{\gamma})}\,\mathrm{d}x' = -k^2\int_{m_{\min}}^{m_{\max}}(x')^{-\alpha}\frac{\gamma(\frac{x'}{M_{\text{turn}}})^{\gamma}\mathrm{e}^{-(\frac{x'}{M_{\text{turn}}})^{\gamma}}}{M_{\text{turn}}}\,\mathrm{d}x'\,.$$

The derivative of k with respect to γ is given by

$$\frac{\partial k(\gamma)}{\partial \gamma} = -k^2\int_{m_{\min}}^{m_{\max}}\frac{\partial}{\partial \gamma}(x')^{-\alpha}\cdot \mathrm{e}^{-((\frac{x}{M_{\text{turn}}})^{\gamma})}\,\mathrm{d}x'\,,$$

with $\mathrm{e}^{-((\frac{x}{M_{\text{turn}}})^{\gamma})} = (\mathrm{e}^{(\frac{x}{M_{\text{turn}}})^{\gamma}})^{-1} = (\mathrm{e}^{\mathrm{e}^{\ln(\frac{x}{M_{\text{turn}}})^{\gamma}}})^{-1}$ from this follows

$$\begin{aligned}
\frac{\partial}{\partial \gamma}\mathrm{e}^{-((\frac{x}{M_{\text{turn}}})^{\gamma})} &= \frac{\partial}{\partial \gamma}(\mathrm{e}^{\mathrm{e}^{\ln(\frac{x}{M_{\text{turn}}})^{\gamma}}})^{-1} = -1\cdot(\mathrm{e}^{\mathrm{e}^{\ln(\frac{x}{M_{\text{turn}}})^{\gamma}}})^{-2}\frac{\partial}{\partial \gamma}(\mathrm{e}^{\mathrm{e}^{\ln(\frac{x}{M_{\text{turn}}})^{\gamma}}})\\
&= \frac{-1}{(\mathrm{e}^{(\frac{x}{M_{\text{turn}}})^{\gamma}})^{\cancel{2}}}\cancel{\mathrm{e}^{(\frac{x}{M_{\text{turn}}})^{\gamma}}}\frac{\partial}{\partial \gamma}(\mathrm{e}^{\ln(\frac{x}{M_{\text{turn}}})\gamma}) = \frac{-1}{(\mathrm{e}^{(\frac{x}{M_{\text{turn}}})^{\gamma}})}\ln(\frac{x}{M_{\text{turn}}})\mathrm{e}^{\ln(\frac{x}{M_{\text{turn}}})\gamma}\\
&= \frac{-1}{(\mathrm{e}^{(\frac{x}{M_{\text{turn}}})^{\gamma}})}\ln(\frac{x}{M_{\text{turn}}})(\frac{x}{M_{\text{turn}}})^{\gamma}
\end{aligned}$$

Therefore the final term is

$$\begin{aligned}
\frac{\partial k(\gamma)}{\partial \gamma} &= -k^2\int_{m_{\min}}^{m_{\max}}(x')^{-\alpha}(-1(\frac{x'}{M_{\text{turn}}})^{\gamma}\ln(\frac{x}{M_{\text{turn}}})\mathrm{e}^{-((\frac{x}{M_{\text{turn}}})^{\gamma})})\,\mathrm{d}x'\\
&= k^2\int_{m_{\min}}^{m_{\max}}(x')^{-\alpha}(\frac{x'}{M_{\text{turn}}})^{\gamma}\ln(\frac{x}{M_{\text{turn}}})\mathrm{e}^{-((\frac{x}{M_{\text{turn}}})^{\gamma})}\,\mathrm{d}x'\,.
\end{aligned}$$

2.3.1 Error calculation

The second derivatives of the log-likelihood function for the modified Schechter function MLE, needed to calculate an error estimate, are determined below.

2.3.2 Second derivatives with respect to α

First, the second derivatives of the log-likelihood function with respect to α are derived.
This yields

$$\frac{\partial^2 \ln \mathcal{L}}{\partial \alpha^2} = n \frac{\partial}{\partial \alpha}\left(\frac{\frac{\partial k(\alpha)}{\partial \alpha}}{k}\right) - 0 = n \frac{\frac{\partial^2 k}{\partial \alpha^2} k - \left(\frac{\partial k(\alpha)}{\partial \alpha}\right)^2}{k^2}$$

$$\frac{\partial}{\partial \alpha}\frac{\partial \ln \mathcal{L}}{\partial M_{\text{turn}}} = n \frac{\partial}{\partial \alpha}\left(\frac{\frac{\partial k(M_{\text{turn}})}{\partial M_{\text{turn}}}}{k}\right) + 0 = n \frac{\frac{\partial}{\partial \alpha}\left(\frac{\partial k}{\partial M_{\text{turn}}}\right) k - \frac{\partial k(\alpha)}{\partial M_{\text{turn}}} \frac{\partial k(\alpha)}{\partial \alpha}}{k^2}$$

$$\frac{\partial}{\partial \alpha}\frac{\partial \ln \mathcal{L}}{\partial \gamma} = n \frac{\partial}{\partial \alpha}\left(\frac{\frac{\partial k(\gamma)}{\partial \gamma}}{k}\right) - 0 = n \frac{\frac{\partial}{\partial \alpha}\left(\frac{\partial k}{\partial \gamma}\right) k - \frac{\partial k(\alpha)}{\partial \gamma} \frac{\partial k(\alpha)}{\partial \alpha}}{k^2}$$

Now, the second derivatives of k with respect to α have to be determined. They are given by

$$\begin{aligned}
\frac{\partial^2 k}{\partial \alpha^2} &= \frac{\partial}{\partial \alpha}\left(k^2 \int_{m_{\min}}^{m_{\max}} \ln(x')(x')^{-\alpha} \cdot \mathrm{e}^{-\left(\left(\frac{x}{M_{\text{turn}}}\right)^{\gamma}\right)} \,\mathrm{d}x'\right) \\
&= \frac{\partial}{\partial \alpha}(k^2) \int_{m_{\min}}^{m_{\max}} (...) dx' + k^2 \frac{\partial}{\partial \alpha} \int_{m_{\min}}^{m_{\max}} \ln(x')(x')^{-\alpha} \cdot \mathrm{e}^{-\left(\left(\frac{x}{M_{\text{turn}}}\right)^{\gamma}\right)} \,\mathrm{d}x' \\
&= 2k \frac{\partial k}{\partial \alpha} \int_{m_{\min}}^{m_{\max}} (...) dx' - k^2 \int_{m_{\min}}^{m_{\max}} (\ln(x'))^2 (x')^{-\alpha} \cdot \mathrm{e}^{-\left(\left(\frac{x}{M_{\text{turn}}}\right)^{\gamma}\right)} \,\mathrm{d}x'
\end{aligned}$$

$$\begin{aligned}
\frac{\partial}{\partial \alpha}\frac{\partial k}{\partial M_{\text{turn}}} &= \frac{\partial}{\partial \alpha}\left(-k^2 \int_{m_{\min}}^{m_{\max}} (x')^{-\alpha} \frac{\gamma\left(\frac{x'}{M_{\text{turn}}}\right)^{\gamma} \mathrm{e}^{-\left(\frac{x'}{M_{\text{turn}}}\right)^{\gamma}}}{M_{\text{turn}}} \,\mathrm{d}x'\right) \\
&= -2k \frac{\partial k}{\partial \alpha} \int_{m_{\min}}^{m_{\max}} (...) dx' + k^2 \int_{m_{\min}}^{m_{\max}} \ln(x')(x')^{-\alpha} \frac{\gamma\left(\frac{x'}{M_{\text{turn}}}\right)^{\gamma} \mathrm{e}^{-\left(\frac{x'}{M_{\text{turn}}}\right)^{\gamma}}}{M_{\text{turn}}} \,\mathrm{d}x'
\end{aligned}$$

$$\begin{aligned}
\frac{\partial}{\partial \alpha}\frac{\partial k}{\partial \gamma} &= \frac{\partial}{\partial \alpha}\left(k^2 \int_{m_{\min}}^{m_{\max}} (x')^{-\alpha} \left(\frac{x'}{M_{\text{turn}}}\right)^{\gamma} \ln\left(\frac{x}{M_{\text{turn}}}\right) \mathrm{e}^{-\left(\left(\frac{x}{M_{\text{turn}}}\right)^{\gamma}\right)} \,\mathrm{d}x'\right) \\
&= \frac{\partial}{\partial \alpha}(k^2) \int_{m_{\min}}^{m_{\max}} (...) dx' + k^2 \frac{\partial}{\partial \alpha} \int_{m_{\min}}^{m_{\max}} (x')^{-\alpha} \left(\frac{x'}{M_{\text{turn}}}\right)^{\gamma} \ln\left(\frac{x}{M_{\text{turn}}}\right) \mathrm{e}^{-\left(\left(\frac{x}{M_{\text{turn}}}\right)^{\gamma}\right)} \,\mathrm{d}x' \\
&= 2k \frac{\partial k}{\partial \alpha} \int_{m_{\min}}^{m_{\max}} (...) dx' - k^2 \int_{m_{\min}}^{m_{\max}} \ln(x')(x')^{-\alpha} \left(\frac{x'}{M_{\text{turn}}}\right)^{\gamma} \ln\left(\frac{x}{M_{\text{turn}}}\right) \mathrm{e}^{-\left(\left(\frac{x}{M_{\text{turn}}}\right)^{\gamma}\right)} \,\mathrm{d}x'
\end{aligned}$$

Note that if the derivative is applied to the function under the interval, it will just lead to a multiplication with $-\ln(x')$ under the previous integral.

2.3.3 Second derivatives with respect to M_{turn}

The second derivatives of the log-likelihood function with respect to M_{turn} yield

$$\frac{\partial}{\partial M_{\text{turn}}}\frac{\partial \ln \mathcal{L}}{\partial \alpha} = n\frac{\partial}{\partial M_{\text{turn}}}\frac{\frac{\partial k(\alpha)}{\partial \alpha}}{k} - 0 = n\frac{\frac{\partial}{\partial M_{\text{turn}}}(\frac{\partial k(\alpha)}{\partial \alpha})k - \frac{\partial k(\alpha)}{\partial \alpha}\frac{\partial k(\alpha)}{\partial M_{\text{turn}}}}{k^2}$$

$$\begin{aligned}\frac{\partial^2 \ln \mathcal{L}}{\partial M_{\text{turn}}^2} &= n\frac{\partial}{\partial M_{\text{turn}}}\frac{\frac{\partial k(M_{\text{turn}})}{\partial M_{\text{turn}}}}{k} - \gamma(\gamma+1)M_{\text{turn}}^{-(\gamma+2)}\sum_{i=1}^{n}(x_i)^{\gamma} \\ &= n\frac{\frac{\partial^2 k}{\partial M_{\text{turn}}^2}k - (\frac{\partial k(M_{\text{turn}})}{\partial M_{\text{turn}}})^2}{k^2} - \gamma(\gamma+1)M_{\text{turn}}^{-(\gamma+2)}\sum_{i=1}^{n}(x_i)^{\gamma}\end{aligned}$$

$$\begin{aligned}\frac{\partial}{\partial M_{\text{turn}}}\frac{\partial \ln \mathcal{L}}{\partial \gamma} &= n\frac{\partial}{\partial M_{\text{turn}}}\frac{\frac{\partial k(\gamma)}{\partial \gamma}}{k} - \frac{\partial}{\partial M_{\text{turn}}}(\sum_{i=1}^{n}\ln(\frac{x_i}{M_{\text{turn}}})(\frac{x_i}{M_{\text{turn}}})^{\gamma}) \\ &= n\frac{\frac{\partial}{\partial M_{\text{turn}}}(\frac{\partial k(\gamma)}{\partial \gamma})k - \frac{\partial k(\gamma)}{\partial \gamma}\frac{\partial k(\gamma)}{\partial M_{\text{turn}}}}{k^2} - \sum_{i=1}^{n}\frac{\partial}{\partial M_{\text{turn}}}(\ln(\frac{x_i}{M_{\text{turn}}})(\frac{x_i}{M_{\text{turn}}})^{\gamma}) \\ &= n\frac{\frac{\partial}{\partial M_{\text{turn}}}(\frac{\partial k(\gamma)}{\partial \gamma})k - \frac{\partial k(\gamma)}{\partial \gamma}\frac{\partial k(\gamma)}{\partial M_{\text{turn}}}}{k^2} - \sum_{i=1}^{n}(-\frac{(\gamma\ln(\frac{x_i}{M_{\text{turn}}})+1)\cdot(\frac{x_i}{M_{\text{turn}}})^{\gamma}}{M_{\text{turn}}}) \\ &= n\frac{\frac{\partial}{\partial M_{\text{turn}}}(\frac{\partial k(\gamma)}{\partial \gamma})k - \frac{\partial k(\gamma)}{\partial \gamma}\frac{\partial k(\gamma)}{\partial M_{\text{turn}}}}{k^2} + \sum_{i=1}^{n}\frac{(\gamma\ln(\frac{x_i}{M_{\text{turn}}})+1)\cdot(\frac{x_i}{M_{\text{turn}}})^{\gamma}}{M_{\text{turn}}}\end{aligned}$$

the corresponding derivatives of k are[2]

$$\begin{aligned}\frac{\partial}{\partial M_{\text{turn}}}\frac{\partial k(\alpha)}{\partial \alpha} &= \frac{\partial}{\partial M_{\text{turn}}}(k^2\int_{m_{\min}}^{m_{\max}}\ln(x')(x')^{-\alpha}\cdot \mathrm{e}^{-((\frac{x}{M_{\text{turn}}})^{\gamma})}\,\mathrm{d}x') \\ &= 2k\frac{\partial k}{\partial M_{\text{turn}}}(\int_{m_{\min}}^{m_{\max}}...dx') + k^2\int_{m_{\min}}^{m_{\max}}\ln(x')(x')^{-\alpha}\cdot\frac{\partial}{\partial M_{\text{turn}}}\mathrm{e}^{-((\frac{x}{M_{\text{turn}}})^{\gamma})}\,\mathrm{d}x' \\ &= 2k\frac{\partial k}{\partial M_{\text{turn}}}(\int_{m_{\min}}^{m_{\max}}...dx') + k^2\int_{m_{\min}}^{m_{\max}}\ln(x')(x')^{-\alpha}\frac{\gamma(\frac{x'}{M_{\text{turn}}})^{\gamma}\mathrm{e}^{-(\frac{x'}{M_{\text{turn}}})^{\gamma}}}{M_{\text{turn}}}\,\mathrm{d}x'\end{aligned}$$

[2]The symbol (...) in the following means that several steps were skipped here. This is mostly due to the derivations being long and tedious, but the result can be confirmed with any derivative calculator.

$$\frac{\partial^2 k}{\partial M_{\text{turn}}^2} = \frac{\partial}{\partial M_{\text{turn}}}\frac{\partial k}{\partial M_{\text{turn}}} = \frac{\partial}{\partial M_{\text{turn}}}\Big(-k^2\int_{m_{\min}}^{m_{\max}}(x')^{-\alpha}\frac{\gamma(\frac{x'}{M_{\text{turn}}})^{\gamma}\mathrm{e}^{-(\frac{x'}{M_{\text{turn}}})^{\gamma}}}{M_{\text{turn}}}\,\mathrm{d}x'\Big)$$

$$= -2k\frac{\partial k}{\partial M_{\text{turn}}}\int_{m_{\min}}^{m_{\max}}(...dx') - k^2\int_{m_{\min}}^{m_{\max}}(x')^{-\alpha}\frac{\partial}{\partial M_{\text{turn}}}\Big(\frac{\gamma(\frac{x'}{M_{\text{turn}}})^{\gamma}\mathrm{e}^{-(\frac{x'}{M_{\text{turn}}})^{\gamma}}}{M_{\text{turn}}}\Big)\,\mathrm{d}x'$$

$$\overset{(...)}{=} -2k\frac{\partial k}{\partial M_{\text{turn}}}\int_{m_{\min}}^{m_{\max}}(...dx') - k^2\int_{m_{\min}}^{m_{\max}}(x')^{-\alpha}\frac{\gamma\Big(\gamma\cdot(\frac{x'}{M_{\text{turn}}})^{\gamma}-\gamma-1\Big)\cdot(\frac{x'}{M_{\text{turn}}})^{\gamma}\mathrm{e}^{-(\frac{x'}{M_{\text{turn}}})^{\gamma}}}{M_{\text{turn}}^2}\,\mathrm{d}x'$$

$$\frac{\partial}{\partial M_{\text{turn}}}\frac{\partial k(\gamma)}{\partial \gamma} = \frac{\partial}{\partial M_{\text{turn}}}\Big(k^2\int_{m_{\min}}^{m_{\max}}(x')^{-\alpha}(\frac{x'}{M_{\text{turn}}})^{\gamma}\ln(\frac{x}{M_{\text{turn}}})\mathrm{e}^{-((\frac{x}{M_{\text{turn}}})^{\gamma})}\,\mathrm{d}x'\Big)$$

$$= 2k\frac{\partial k}{\partial M_{\text{turn}}}\int_{m_{\min}}^{m_{\max}}(...dx') + k^2\int_{m_{\min}}^{m_{\max}}(x')^{-\alpha}\frac{\partial}{\partial M_{\text{turn}}}\Big((\frac{x'}{M_{\text{turn}}})^{\gamma}\ln(\frac{x}{M_{\text{turn}}})\mathrm{e}^{-((\frac{x}{M_{\text{turn}}})^{\gamma})}\Big)\,\mathrm{d}x'$$

$$\overset{(...)}{=} 2k\frac{\partial k}{\partial M_{\text{turn}}}\int_{m_{\min}}^{m_{\max}}(...dx')$$

$$+ k^2\int_{m_{\min}}^{m_{\max}}(x')^{-\alpha}\frac{\Big(\gamma\ln(\frac{x'}{M_{\text{turn}}})\cdot(\frac{x'}{M_{\text{turn}}})^{\gamma}-\gamma\ln(\frac{x'}{M_{\text{turn}}})-1\Big)(\frac{x'}{M_{\text{turn}}})^{\gamma}\mathrm{e}^{-((\frac{x}{M_{\text{turn}}})^{\gamma})}}{M_{\text{turn}}}\,\mathrm{d}x'$$

2.3.4 The second derivative with respect to γ

Lastly, the second derivatives of the likelihood function with respect to γ are derived. Recall that

$$\frac{\partial \ln\mathcal{L}}{\partial\alpha} = n\frac{\frac{\partial k(\alpha)}{\partial\alpha}}{k} - \sum_{i=1}^{n}(\ln(x_i))$$

$$\Rightarrow \frac{\partial}{\partial\gamma}\frac{\partial\ln\mathcal{L}}{\partial\alpha} = n\frac{\partial}{\partial\gamma}\frac{\frac{\partial k(\alpha)}{\partial\alpha}}{k} - 0 = n\frac{\frac{\partial}{\partial\gamma}(\frac{\partial k(\alpha)}{\partial\alpha})k - \frac{\partial k(\alpha)}{\partial\alpha}\frac{\partial k(\alpha)}{\partial\gamma}}{k^2}$$

Similarly,

$$\frac{\partial \ln\mathcal{L}}{\partial M_{\text{turn}}} = n\frac{\frac{\partial k(M_{\text{turn}})}{\partial M_{\text{turn}}}}{k} + \gamma M_{\text{turn}}^{-(\gamma+1)}\sum_{i=1}^{n}(x_i)^{\gamma}$$

$$\Rightarrow \frac{\partial}{\partial\gamma}\frac{\partial\ln\mathcal{L}}{\partial M_{\text{turn}}} = n\frac{\frac{\partial}{\partial\gamma}(\frac{\partial k(M_{\text{turn}})}{\partial M_{\text{turn}}})k - \frac{\partial k(M_{\text{turn}})}{\partial M_{\text{turn}}}\frac{\partial k(M_{\text{turn}})}{\partial\gamma}}{k^2} + \frac{\partial}{\partial\gamma}(\gamma M_{\text{turn}}^{-(\gamma+1)})\sum_{i=1}^{n}(x_i)^{\gamma} + \gamma M_{\text{turn}}^{-(\gamma+1)}\sum_{i=1}^{n}\frac{\partial}{\partial\gamma}(x_i)^{\gamma}$$

$$= n\frac{\frac{\partial}{\partial\gamma}(\frac{\partial k(M_{\text{turn}})}{\partial M_{\text{turn}}})k - \frac{\partial k(M_{\text{turn}})}{\partial M_{\text{turn}}}\frac{\partial k(M_{\text{turn}})}{\partial\gamma}}{k^2} + (-M_{\text{turn}}^{-(\gamma+1)}(\ln(M_{\text{turn}})\gamma-1))\sum_{i=1}^{n}(x_i)^{\gamma} + \gamma M_{\text{turn}}^{-(\gamma+1)}\sum_{i=1}^{n}\ln(x_i)(x_i)^{\gamma}$$

$$= n\frac{\frac{\partial}{\partial\gamma}(\frac{\partial k(M_{\text{turn}})}{\partial M_{\text{turn}}})k - \frac{\partial k(M_{\text{turn}})}{\partial M_{\text{turn}}}\frac{\partial k(M_{\text{turn}})}{\partial\gamma}}{k^2} - M_{\text{turn}}^{-(\gamma+1)}(\ln(M_{\text{turn}})\gamma-1)\sum_{i=1}^{n}(x_i)^{\gamma} + \gamma M_{\text{turn}}^{-(\gamma+1)}\sum_{i=1}^{n}\ln(x_i)(x_i)^{\gamma}$$

And finally,

$$\frac{\partial \ln \mathcal{L}}{\partial \gamma} = n\frac{\frac{\partial k(\gamma)}{\partial \gamma}}{k} - \sum_{i=1}^{n} \ln(\frac{x_i}{M_{\text{turn}}})(\frac{x_i}{M_{\text{turn}}})^{\gamma}$$

$$\Rightarrow \frac{\partial^2 \ln \mathcal{L}}{\partial \gamma^2} = n\frac{\frac{\partial^2 k}{\partial \gamma^2}k - (\frac{\partial k(\gamma)}{\partial \gamma})^2}{k^2} - \sum_{i=1}^{n} (\ln(\frac{x_i}{M_{\text{turn}}}))^2(\frac{x_i}{M_{\text{turn}}})^{\gamma}$$

The corresponding derivatives of k are:

$$\begin{aligned}
\frac{\partial}{\partial \gamma}(\frac{\partial k(\alpha)}{\partial \alpha}) &= \frac{\partial}{\partial \gamma}(k^2 \int_{m_{\min}}^{m_{\max}} \ln(x')(x')^{-\alpha} \cdot \mathrm{e}^{-((\frac{x'}{M_{\text{turn}}})^{\gamma})}\,\mathrm{d}x') \\
&= 2k\frac{\partial k}{\partial \gamma}\int_{m_{\min}}^{m_{\max}} (\ldots \mathrm{d}x') + k^2 \int_{m_{\min}}^{m_{\max}} \ln(x')(x')^{-\alpha} \cdot \frac{\partial}{\partial \gamma}\mathrm{e}^{-((\frac{x'}{M_{\text{turn}}})^{\gamma})}\,\mathrm{d}x' \\
&= 2k\frac{\partial k}{\partial \gamma}\int_{m_{\min}}^{m_{\max}} (\ldots \mathrm{d}x') + k^2 \int_{m_{\min}}^{m_{\max}} \ln(x')(x')^{-\alpha} \cdot \Big(-(\frac{x'}{M_{\text{turn}}})^{\gamma}\ln(\frac{x'}{M_{\text{turn}}})\mathrm{e}^{-((\frac{x'}{M_{\text{turn}}})^{\gamma})}\Big)\,\mathrm{d}x' \\
&= 2k\frac{\partial k}{\partial \gamma}\int_{m_{\min}}^{m_{\max}} (\ldots \mathrm{d}x') - k^2 \int_{m_{\min}}^{m_{\max}} \ln(x')(x')^{-\alpha} \cdot (\frac{x'}{M_{\text{turn}}})^{\gamma}\ln(\frac{x'}{M_{\text{turn}}})\mathrm{e}^{-((\frac{x'}{M_{\text{turn}}})^{\gamma})}\,\mathrm{d}x'
\end{aligned}$$

$$\begin{aligned}
&\frac{\partial}{\partial \gamma}(\frac{\partial k(M_{\text{turn}})}{\partial M_{\text{turn}}}) = \frac{\partial}{\partial \gamma}(-k^2 \int_{m_{\min}}^{m_{\max}} (x')^{-\alpha}\frac{\gamma(\frac{x'}{M_{\text{turn}}})^{\gamma}\mathrm{e}^{-(\frac{x'}{M_{\text{turn}}})^{\gamma}}}{M_{\text{turn}}}\,\mathrm{d}x') \\
&= -2k\frac{\partial k}{\partial \gamma}\int_{m_{\min}}^{m_{\max}} (\ldots \mathrm{d}x') - k^2 \int_{m_{\min}}^{m_{\max}} (x')^{-\alpha}\frac{\partial}{\partial \gamma}(\frac{\gamma(\frac{x'}{M_{\text{turn}}})^{\gamma}\mathrm{e}^{-(\frac{x'}{M_{\text{turn}}})^{\gamma}}}{M_{\text{turn}}})\,\mathrm{d}x' \\
&\overset{(\ldots)}{=} -2k\frac{\partial k}{\partial \gamma}\int_{m_{\min}}^{m_{\max}} (\ldots \mathrm{d}x') - k^2 \int_{m_{\min}}^{m_{\max}} (x')^{-\alpha}(-\frac{(\frac{x'}{M_{\text{turn}}})^{\gamma}\left(\ln(\frac{x'}{M_{\text{turn}}})\Big((\frac{x'}{M_{\text{turn}}})^{\gamma}-1\Big)\gamma - 1\right)\mathrm{e}^{-(\frac{x'}{M_{\text{turn}}})^{\gamma}}}{M_{\text{turn}}})\,\mathrm{d}x' \\
&= -2k\frac{\partial k}{\partial \gamma}\int_{m_{\min}}^{m_{\max}} (\ldots \mathrm{d}x') + k^2 \int_{m_{\min}}^{m_{\max}} (x')^{-\alpha}\frac{(\frac{x'}{M_{\text{turn}}})^{\gamma}\left(\ln(\frac{x'}{M_{\text{turn}}})\Big((\frac{x'}{M_{\text{turn}}})^{\gamma}-1\Big)\gamma - 1\right)\mathrm{e}^{-(\frac{x'}{M_{\text{turn}}})^{\gamma}}}{M_{\text{turn}}}\,\mathrm{d}x'
\end{aligned}$$

$$\begin{aligned}
\frac{\partial^2 k}{\partial \gamma^2} &= \frac{\partial}{\partial \gamma}(\frac{\partial k(\gamma)}{\partial \gamma}) = \frac{\partial}{\partial \gamma}(k^2 \int_{m_{\min}}^{m_{\max}} (x')^{-\alpha}(\frac{x'}{M_{\text{turn}}})^{\gamma}\ln(\frac{x}{M_{\text{turn}}})\mathrm{e}^{-((\frac{x'}{M_{\text{turn}}})^{\gamma})}\,\mathrm{d}x') \\
&= 2k\frac{\partial k}{\partial \gamma}\int_{m_{\min}}^{m_{\max}} (\ldots \mathrm{d}x') + k^2 \int_{m_{\min}}^{m_{\max}} (x')^{-\alpha}\frac{\partial}{\partial \gamma}\Big((\frac{x'}{M_{\text{turn}}})^{\gamma}\ln(\frac{x}{M_{\text{turn}}})\mathrm{e}^{-((\frac{x'}{M_{\text{turn}}})^{\gamma})}\Big)\,\mathrm{d}x' \\
&\overset{(\ldots)}{=} 2k\frac{\partial k}{\partial \gamma}\int_{m_{\min}}^{m_{\max}} (\ldots \mathrm{d}x') + k^2 \int_{m_{\min}}^{m_{\max}} (x')^{-\alpha}\left(-(\frac{x'}{M_{\text{turn}}})^{\gamma}\ln^2(\frac{x'}{M_{\text{turn}}})\Big((\frac{x'}{M_{\text{turn}}})^{\gamma}-1\Big)\mathrm{e}^{-((\frac{x'}{M_{\text{turn}}})^{\gamma})}\right)\mathrm{d}x' \\
&= 2k\frac{\partial k}{\partial \gamma}\int_{m_{\min}}^{m_{\max}} (\ldots \mathrm{d}x') - k^2 \int_{m_{\min}}^{m_{\max}} (x')^{-\alpha}(\frac{x'}{M_{\text{turn}}})^{\gamma}\ln^2(\frac{x'}{M_{\text{turn}}})\Big((\frac{x'}{M_{\text{turn}}})^{\gamma}-1\Big)\mathrm{e}^{-((\frac{x'}{M_{\text{turn}}})^{\gamma})}\,\mathrm{d}x'
\end{aligned}$$

2.3.5 On possible errors in the derivations

In such long and tedious derivations, a small error can easily sneak in. A way to test whether the error calculation was correct without going through every derivation again is to check whether the Hessian matrix is symmetrical. Since the partial second derivatives of the MS function are continuous on the considered parameter space, the Hessian matrix should be symmetrical. Since this was the case for several datasets and the error calculation of the diagonal entries was checked several times, I am confident that these derivations are correct.

References

Abdey, J. S. 2013, Law, Probability and Risk, 12, 135

Crawford, D. F., Jauncey, D. L., & Murdoch, H. S. 1970, Astrophysical Journal, vol. 162, p. 405, 162, 405

King, G. 1998, Unifying Political Methodology: The Likelihood Theory of Statistical Inference (University of Michigan Press)

Lehmann, E. L., & Casella, G. 1998, Theory of point estimation (Springer)

Lieberz, P., & Kroupa, P. 2017, Monthly Notices of the Royal Astronomical Society, 465, 3775

Maschberger, T., & Kroupa, P. 2009, Monthly Notices of the Royal Astronomical Society, 395, 931

Neyman, J., & Pearson, E. S. 1933, Philosophical Transactions of the Royal Society of London. Series A, Containing Papers of a Mathematical or Physical Character, 231, 289

Scott Long, J. 1997, Advanced quantitative techniques in the social sciences, 7

White, E. P., Enquist, B. J., & Green, J. L. 2008, Ecology, 89, 905

Wilks, S. S. 1938, The annals of mathematical statistics, 9, 60